The Periodic Table of the Elements

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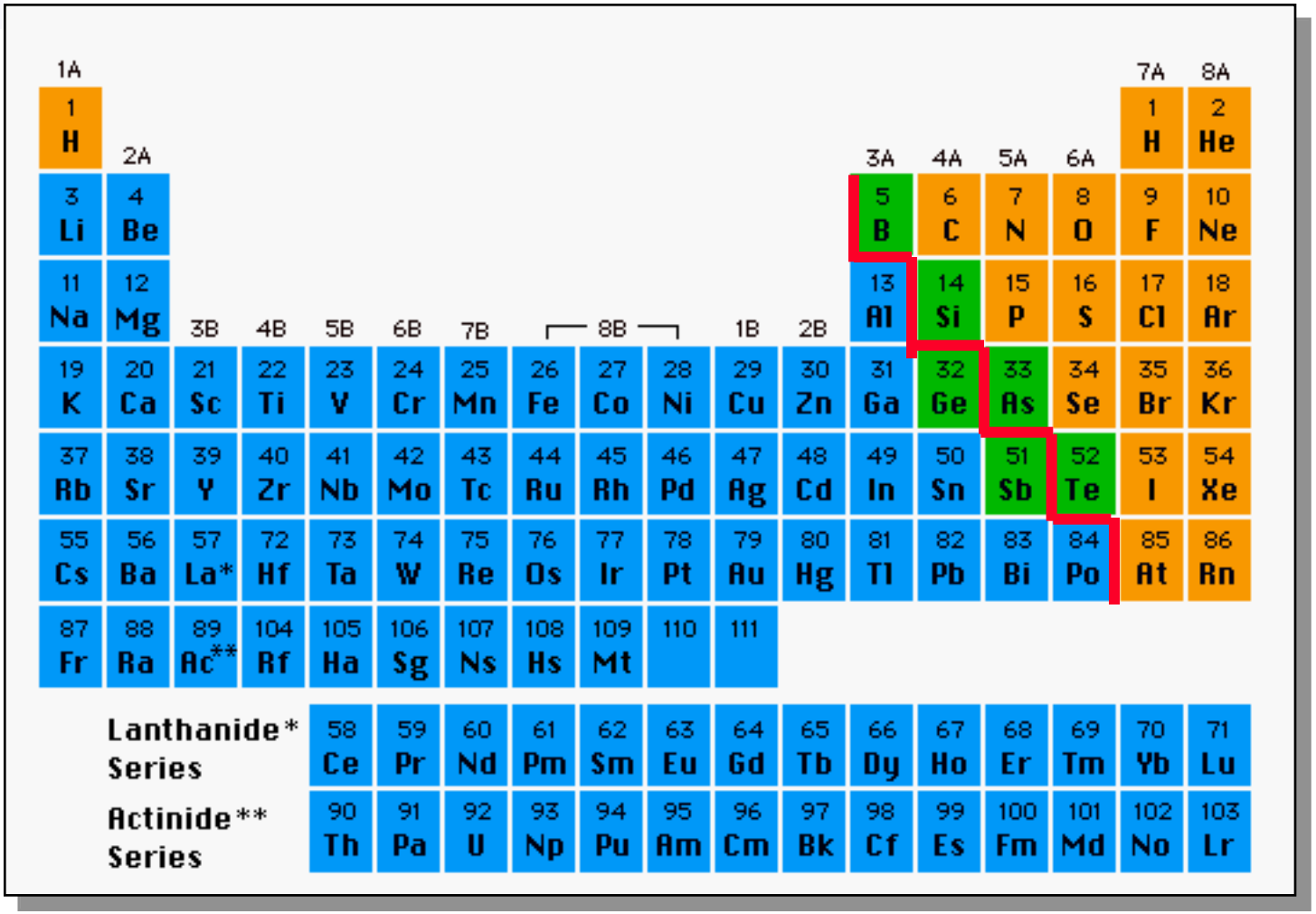
Objectives

- By the end of this lesson, we should be able to:
  - know the electron configurations of atoms and ions up to $Z = 36$ in terms of levels and sub-levels (orbitals) s, p and d
  - know the meaning of the term *ionisation energy*
  - understand how ionisation energies in Period 3 (Na – Ar) and in Group 2 (Be – Ba) give evidence for electron arrangement in sub-levels and in levels
  - classify an element as s, p or d block according to its position in the Periodic Table be able to describe the trends in atomic radius, first ionisation energy, melting and boiling points of the elements Na – Ar
  - understand the reasons for the trends in these properties
  - explain how orbital energies are affected when more than one electron is present in an atom.
  - describe the three basic principles governing electron configuration (Minimization of Energy, Pauli Exclusion Principle, and Hund's rule).
  - apply the Aufbau principle.
  - explain how the Aufbau principle explains the arrangement of the Periodic Table, and vice versa.
  - use the Aufbau principle to explain the nature of the valence electrons and how this
  - defines trends in the Periodic Table.

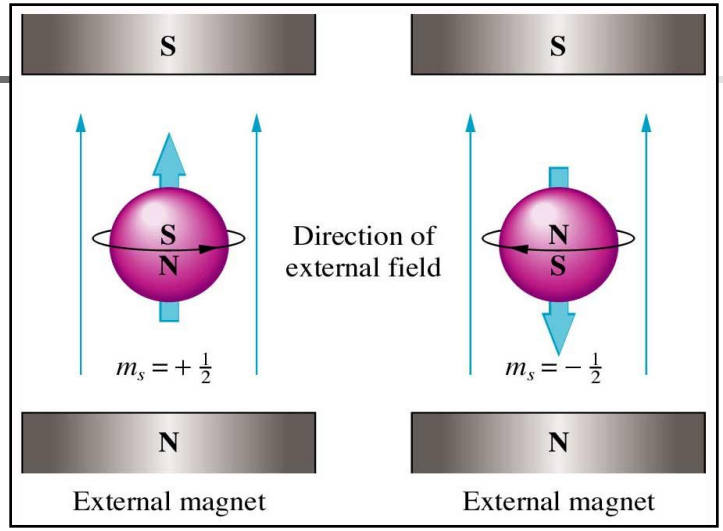
The Modern Periodic Law

- Element properties: periodic function to their atomic numbers.
- The **periodicity** in properties is due to repetition of similar outer shell electronic configuration at certain regular intervals.
- **Periodic table:** based on modern periodic law in which elements are arranged in increasing order of atomic numbers.
- The elements are arranged in **rows [periods]** and **columns [groups]**;
- The table consists of 7 periods and 18 groups
- Period (arabic numerals) indicates the value of 'n' (principal quantum number) for the outermost or valence shell.
- Same number of electrons is present in the outer orbitals (that is, similar valence shell electronic configuration

ATOMIC ELECTRON CONFIGURATIONS AND PERIODICITY

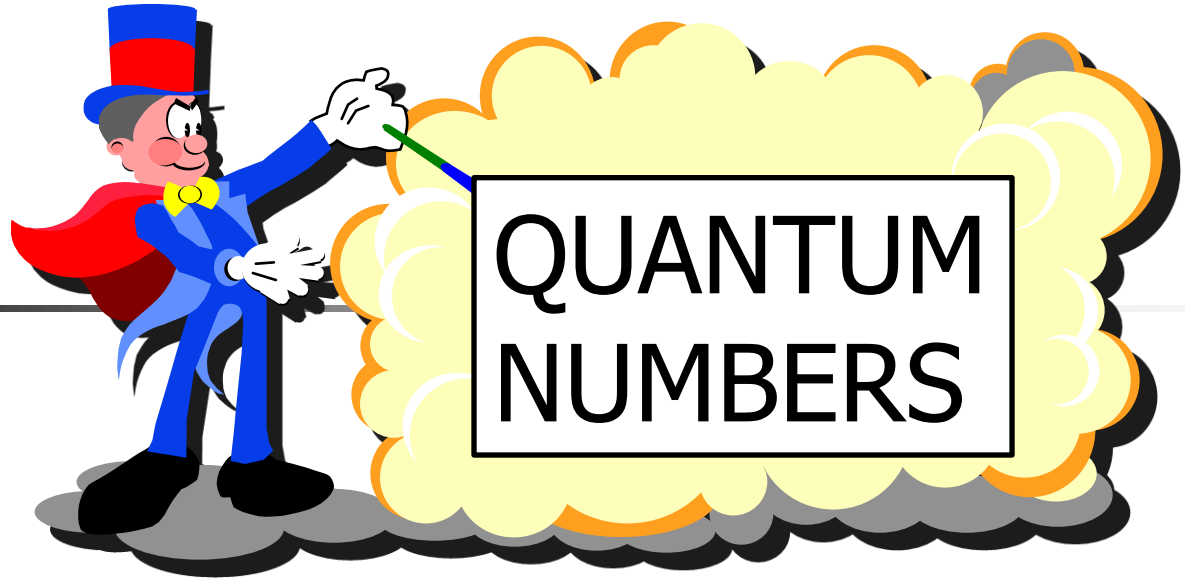


Arrangement of Electrons in Atoms



Electron Spin Quantum Number, m_s

- Each orbital can be assigned no more than 2 electrons!
- This is tied to the existence of a 4th quantum number, the **electron spin quantum number, m_s.**
- Can be proved experimentally that electron has a spin. Two spin directions are given by m_s; where m_s = +1/2 and -1/2.
- **Diamagnetic:** NOT attracted to a magnetic field
- **Paramagnetic:** substance is attracted to a magnetic field (substance has **unpaired electrons)**.



n	--->	shell	1, 2, 3, 4, ...
l	--->	subshell	0, 1, 2, ... n - 1
m_l	--->	orbital	-l ... 0 ... +l
m_s	--->	electron spin	+1/2 and -1/2

Table 8.1 • Number of Electrons Accommodated in Electron Shells and Subshells With $n = 1$ to 6

Electron Shell (n)	Subshells Available	Orbitals Available ($2\ell + 1$)	Number of Electrons Possible in Subshell [$2(2\ell + 1)$]	Maximum Electrons Possible for nth Shell ($2n^2$)
1	*s*	1	2	2
2	*s*	1	2	8
	p	3	6	
3	*s*	1	2	18
	p	3	6	
	d	5	10	
4	*s*	1	2	32
	p	3	6	
	d	5	10	
	f	7	14	
5	*s*	1	2	50
	p	3	6	
	d	5	10	
	f	7	14	
	*g**	9	18	
6	*s*	1	2	72
	p	3	6	
	d	5	10	
	*f**	7	14	
	*g**	9	18	
	*h**	11	22	

*These orbitals are not used in the ground state of any known element.

Electron Configuration

- **"electron configuration"** of an atom is a particular distribution of electrons among available sub shells.
  - The notation for a configuration lists the sub-shell symbols sequentially with a superscript indicating the number of electrons occupying that sub shell.
  - E.g. **Li** (atomic number 3) has 2 electrons in the "1s" sub shell and 1 electron in the "2s" sub shell = $\mathbf{1s^2\ 2s^1}$.
- An **orbital diagram** is used to show how the orbitals of a sub shell are occupied by electrons.
  - Each orbital is represented by a **circle** or **square** and each group of orbitals is labeled by its sub shell notation.
  - Electrons are represented by arrows: up for $m_s = +1/2$ and down for $m_s = -1/2$

Ways of writing configurations

- Spectroscopic/ spdf notation:

spdf notation
for H, atomic number = 1

$1\,s^{1}$

no. of electrons

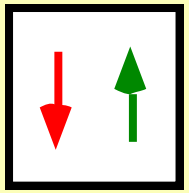
value of n

value of l

- Orbital box notation:

ORBITAL BOX NOTATION
for He, atomic number = 2

$1\,s^{2}$



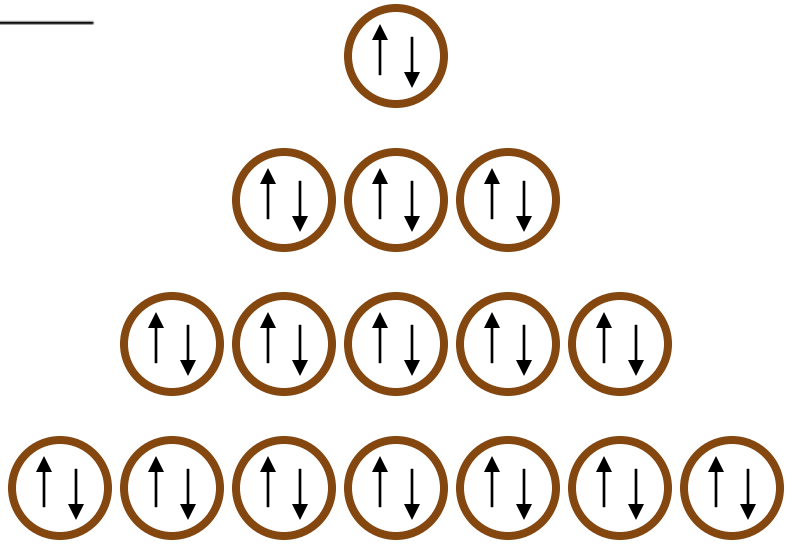
1s

Arrows depict electron spin

Rules for electron configuration

- **<u>Pauli's Exclusion principle:</u>** An orbital can contain a maximum number of 2 electrons and these two electrons must be of opposite spin.

Sub shell	Number of Orbitals	Maximum Number of Electrons
s ($l = 0$)	1	2
p ($l = 1$)	3	6
d ($l = 2$)	5	10
f ($l = 3$)	7	14



- **<u>Aufbau principle:</u>** Every atom has an infinite number of possible electron configurations.
  - **Ground state:** Lowest energy level configuration of atom.
  - Other configurations correspond to **"excited states."**
- The **Aufbau principle** is a scheme used to reproduce the ground state electron configurations of atoms by following the "building up" order.
  - Listed below is the order in which all the possible sub-shells fill with electrons: **1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f.**
  - **No need to memorize the order; we have an easy way!!**

Order for Filling Atomic Subshells

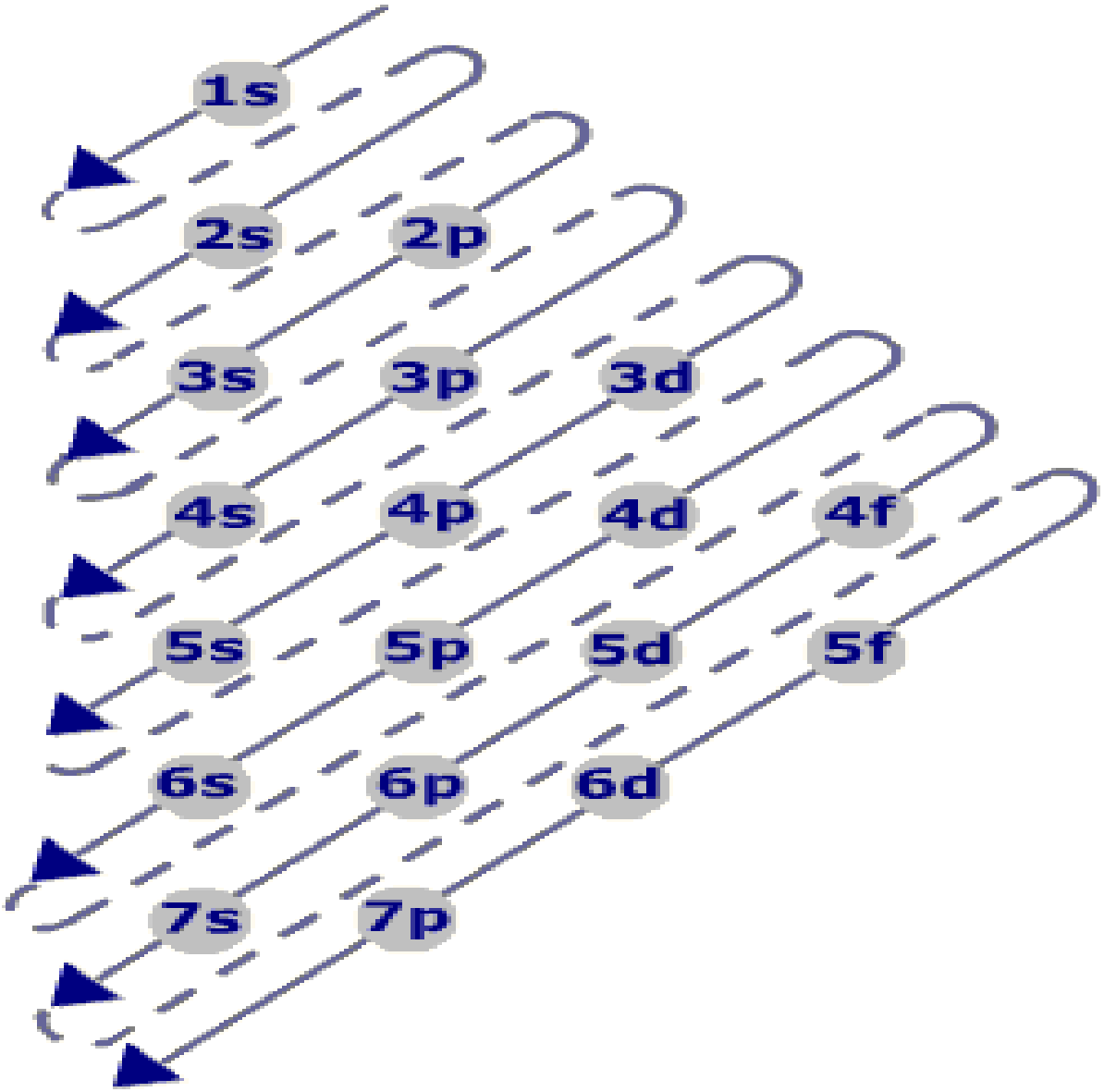


Table 8.1

Ground-State Electron Configurations of Atoms Z = 1 to 36*

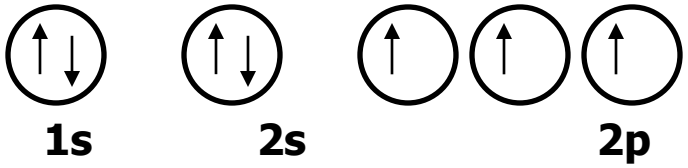
Z	Element	Configuration	Z	Element	Configuration
1	H	$1s^1$	19	K	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$
2	He	$1s^2$	20	Ca	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
3	Li	$1s^2 2s^1$	21	Sc	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^1 4s^2$
4	Be	$1s^2 2s^2$	22	Ti	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^2 4s^2$
5	B	$1s^2 2s^2 2p^1$	23	V	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^3 4s^2$
6	C	$1s^2 2s^2 2p^2$	24	Cr	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$
7	N	$1s^2 2s^2 2p^3$	25	Mn	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$
8	O	$1s^2 2s^2 2p^4$	26	Fe	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
9	F	$1s^2 2s^2 2p^5$	27	Co	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
10	Ne	$1s^2 2s^2 2p^6$	28	Ni	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$
11	Na	$1s^2 2s^2 2p^6 3s^1$	29	Cu	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$
12	Mg	$1s^2 2s^2 2p^6 3s^2$	30	Zn	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2$
13	Al	$1s^2 2s^2 2p^6 3s^2 3p^1$	31	Ga	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^1$
14	Si	$1s^2 2s^2 2p^6 3s^2 3p^2$	32	Ge	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^2$
15	P	$1s^2 2s^2 2p^6 3s^2 3p^3$	33	As	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^3$
16	S	$1s^2 2s^2 2p^6 3s^2 3p^4$	34	Se	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^4$
17	Cl	$1s^2 2s^2 2p^6 3s^2 3p^5$	35	Br	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^5$
18	Ar	$1s^2 2s^2 2p^6 3s^2 3p^6$	36	Kr	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$

*A complete table appears in Appendix D.

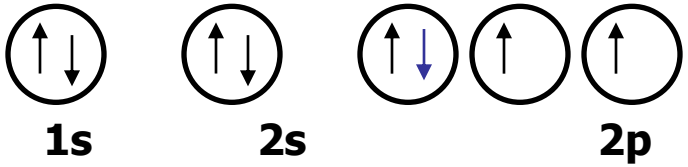
- **<u>Hund's rule of maximum multiplicity:</u>** Electron pairing in p, d and f orbital cannot occur until each orbital of a given subshell contains one electron each or is singly occupied".
  - **Exceptional Configurations**: **half filled & fully filled orbitals**
    - Cu = 29 electrons; electronic configuration is: **$1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^2, 3d^9$**
    - But 1 electron shifts from lower energy 4s orbital to higher energy 3d orbital
    - Thus electronic configuration of:

      Cu = $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1, 3d^{10}$
  - Therefore, fully filled & half filled orbitals are more stable.

- To apply Hund's rule to oxygen, whose ground state configuration is **$1s^2 2s^2 2p^4$,** we place the first seven electrons as follows.



  - The last electron is paired with one of the 2p electrons to give a doubly occupied orbital.



Magnetic Properties

- Although an electron behaves like a tiny magnet, two electrons that are opposite in spin cancel each other. Only atoms with unpaired electrons exhibit magnetic susceptibility.
  - A paramagnetic substance is one that is weakly attracted by a magnetic field, usually the result of unpaired electrons.
  - A diamagnetic substance is not attracted by a magnetic field generally because it has only paired electrons.

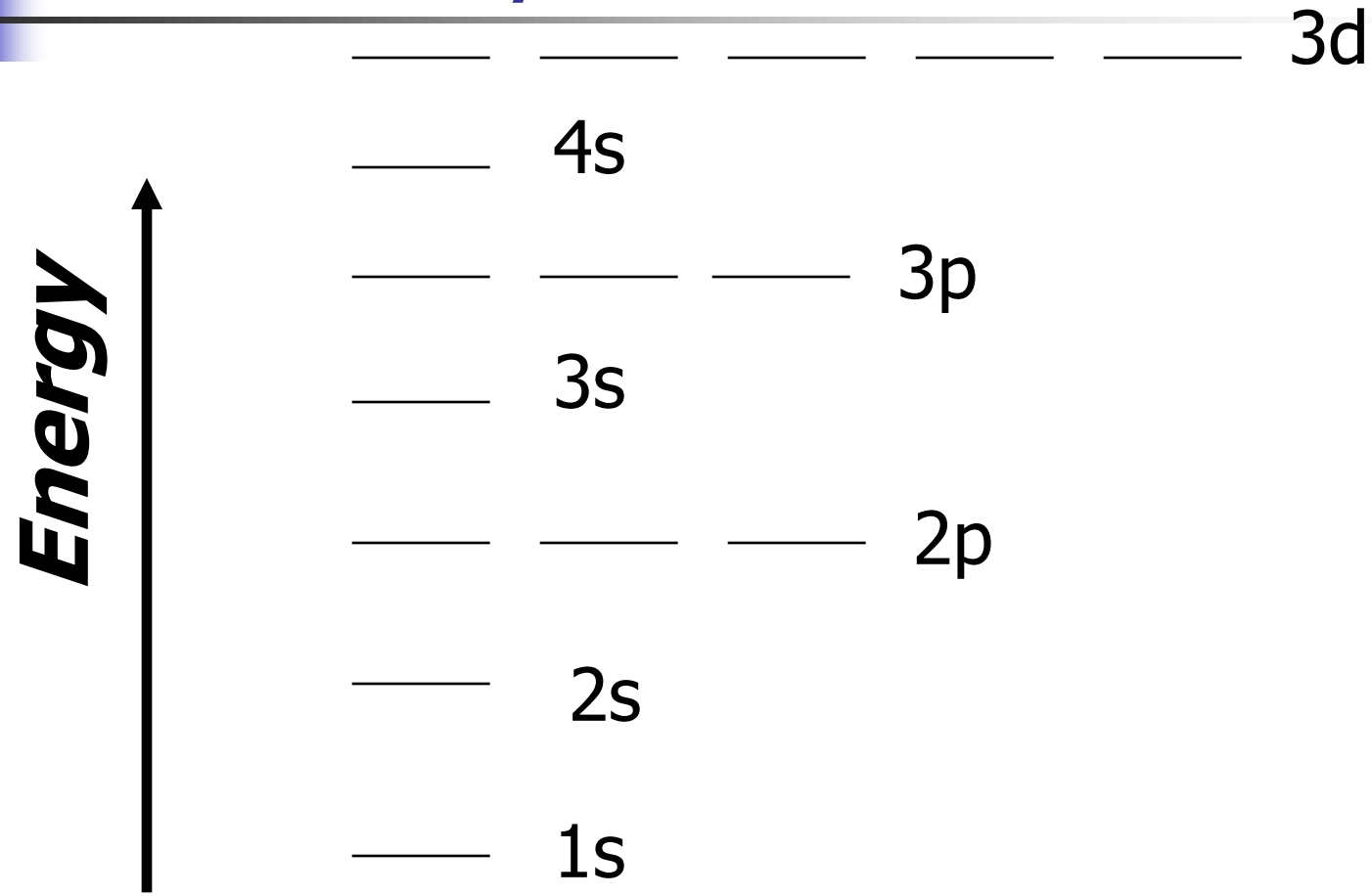
Which are paramagnetic or diamagnetic?

Table 8.2

Orbital Diagrams for the Ground States of Atoms from $Z = 1$ to $Z = 10$

Atom	Z	Configuration	Orbital Diagram 1s	2s	2p
Hydrogen	1	$1s^1$	↑	○	○○○
Helium	2	$1s^2$	↑↓	○	○○○
Lithium	3	$1s^2 2s^1$	↑↓	↑	○○○
Beryllium	4	$1s^2 2s^2$	↑↓	↑↓	○○○
Boron	5	$1s^2 2s^2 2p^1$	↑↓	↑↓	↑○○
Carbon	6	$1s^2 2s^2 2p^2$	↑↓	↑↓	↑↑○
Nitrogen	7	$1s^2 2s^2 2p^3$	↑↓	↑↓	↑↑↑
Oxygen	8	$1s^2 2s^2 2p^4$	↑↓	↑↓	↑↓↑↑
Fluorine	9	$1s^2 2s^2 2p^5$	↑↓	↑↓	↑↓↑↓↑
Neon	10	$1s^2 2s^2 2p^6$	↑↓	↑↓	↑↓↑↓↑↓

Orbital Energy Levels in Multi-electron Systems

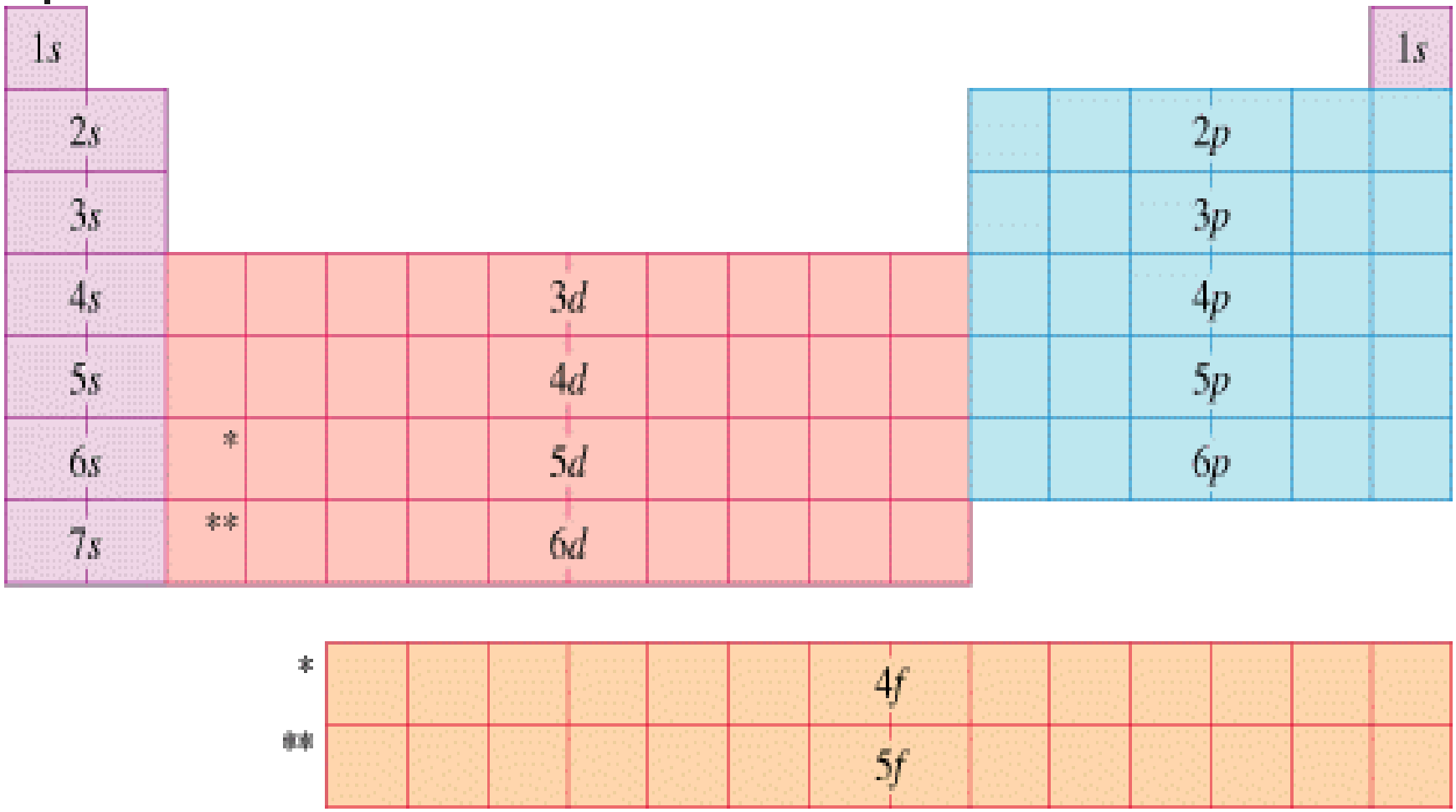


1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p

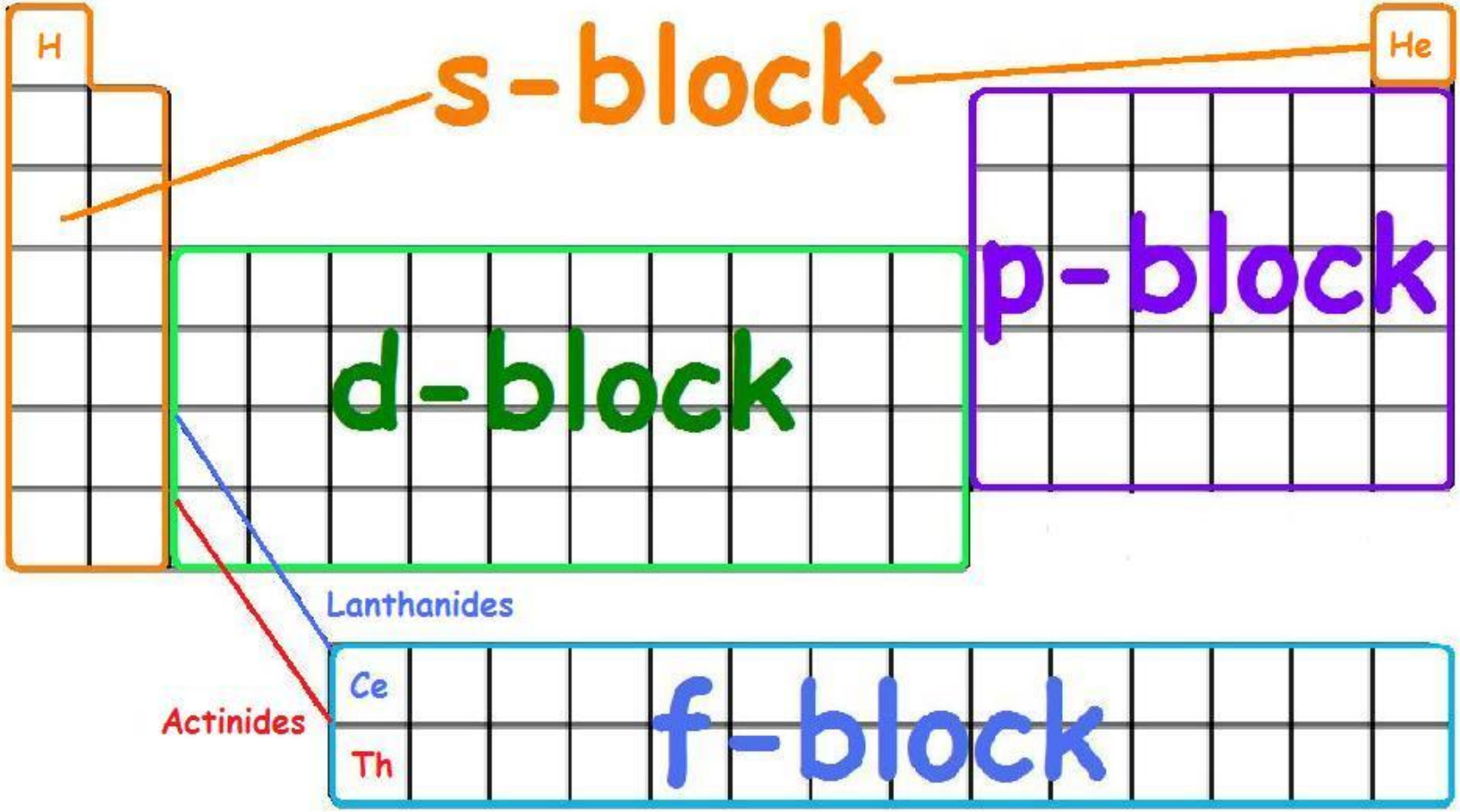
Configurations and the Periodic Table

- The following slide illustrates how the periodic table provides a sound way to remember the Aufbau sequence. In many cases you need only the configuration of the outer electrons.
  - You can determine this from their position on the periodic table.
  - The total number of valence electrons for an atom equals its group number.

Configurations and the Periodic Table



Classification of Elements



Aufbau Principle

- Here are a few examples. Using the abbreviation [He] for $1s^2$, the configurations are

Z=3	**Lithium**	$1s^2 2s^1$	**or**	$[He]2s^1$
Z=4	**Beryllium**	$1s^2 2s^2$	**or**	$[He]2s^2$

- With boron (Z=5), the electrons begin filling the 2p subshell.

Z=5	**Boron**	$1s^2 2s^2 2p^1$	**or**	$[He]2s^2 2p^1$
Z=6	**Carbon**	$1s^2 2s^2 2p^2$	**or**	$[He]2s^2 2p^2$
Z=7	**Nitrogen**	$1s^2 2s^2 2p^3$	**or**	$[He]2s^2 2p^3$
Z=8	**Oxygen**	$1s^2 2s^2 2p^4$	**or**	$[He]2s^2 2p^4$

Aufbau Principle

- With sodium (Z = 11), the 3s sub shell begins to fill.

Z=11	**Sodium**	**$1s^2 2s^2 2p^6 3s^1$**	**or**	**$[Ne]3s^1$**
Z=12	**Magnesium**	**$1s^2 2s^2 2p^2 3s^2$**	**or**	**$[Ne]3s^2$**

  - Then the 3p sub shell begins to fill.

Z=13	**Aluminum**	**$1s^2 2s^2 2p^6 3s^2 3p^1$**	**or**	**$[Ne]3s^2 3p^1$**

-
-

Z=18	**Argon**	**$1s^2 2s^2 2p^6 3s^2 3p^6$**	**or**	**$[Ne]3s^2 3p^6$ or [Ar]**

Configurations and the Periodic Table

- Note that elements within a given family have similar configurations.
  - For instance, look at the ***noble gases***.

Helium	$1s^2$
Neon	$1s^2\underline{\mathbf{2s^2 2p^6}}$
Argon	$1s^2 2s^2 2p^6\underline{\mathbf{3s^2 3p^6}}$
Krypton	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}\underline{\mathbf{4s^2 4p^6}}$

  - The Group IIA elements (***alkaline earth metals)***.

Beryllium	$1s^2\underline{\mathbf{2s^2}}$
Magnesium	$1s^2 2s^2 2p^6\underline{\mathbf{3s^2}}$
Calcium	$1s^2 2s^2 2p^6 3s^2 3p^6\underline{\mathbf{4s^2}}$

Configurations and the Periodic Table

- Electrons that ***reside in the outermost shell*** of an atom - or in other words, those electrons outside the "noble gas core"- are called **valence electrons**.
  - These electrons are primarily involved in chemical reactions.
  - Elements within a given group have the same "valence shell configuration."
  - This accounts for the similarity of the chemical properties among groups of elements.

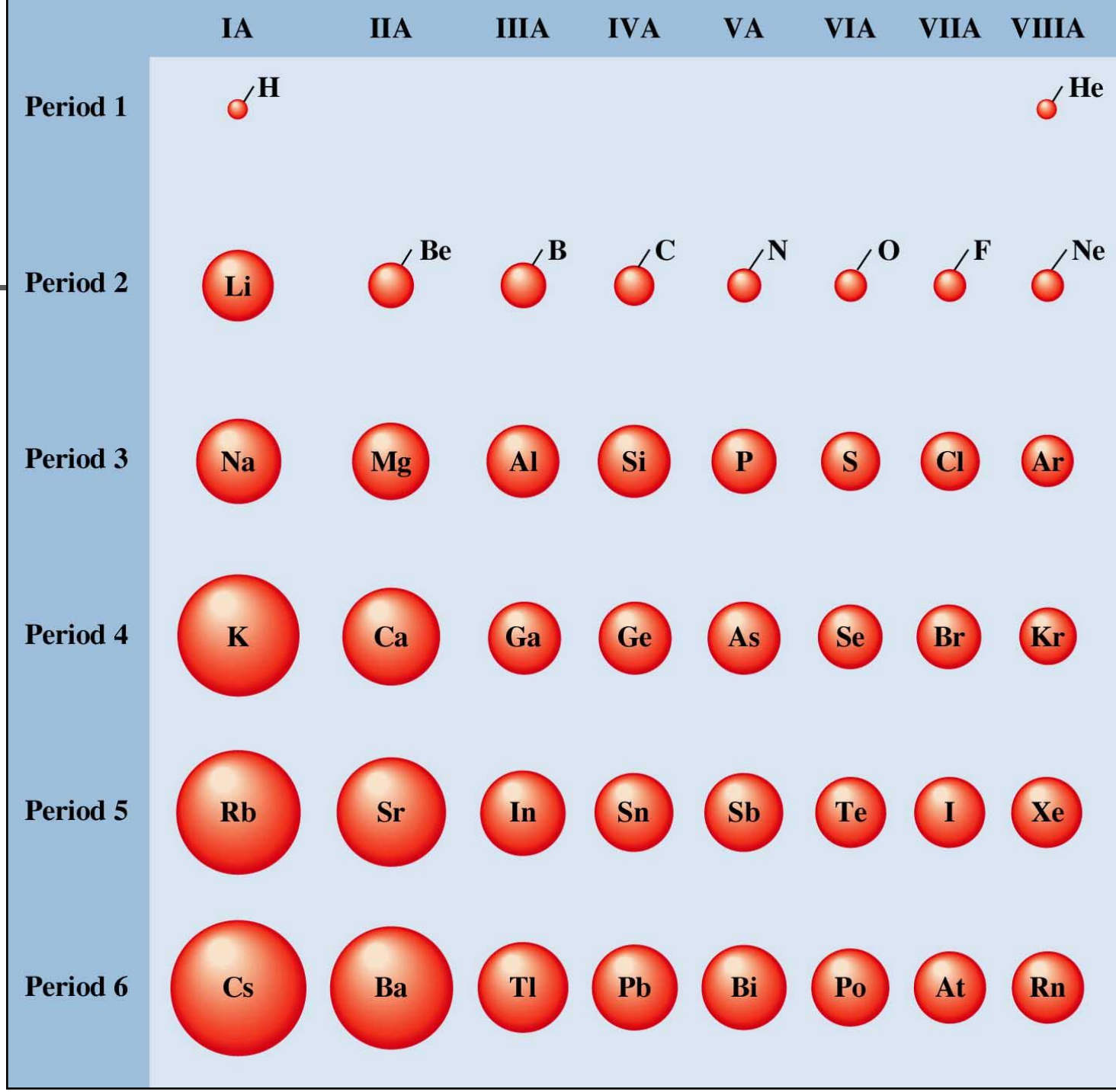
Periodic Properties

- The **periodic law** states that ***when the elements are arranged by atomic number, their physical and chemical properties vary periodically***.
- We will look at three periodic properties:
  - **Atomic and ionic size (radius)**
  - **Ionization energy**
  - **Electron affinity**

Atomic Radius

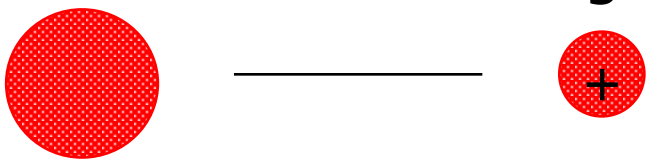
- **Within each period:** the atomic radius tends to decrease with increasing atomic number (nuclear charge).
- **Within each group:** the atomic radius tends to increase with the period number.
- Two factors determine the size of an atom.
  - **Principal quantum number (n):** The larger is "n", the larger the size of the orbital.
  - **Effective nuclear charge:** which is the positive charge an electron experiences from the nucleus minus any "shielding effects" from intervening electrons.

Representation of atomic radii (covalent radii) of the main-group elements.



Ion size

- What happens to atomic size when it loses electron?
  - CATIONS are SMALLER than their parent atoms.
  - The electron/proton attraction has gone UP and so size DECREASES.



Li,152 pm
3e and 3p

Li $^{+}$, 78 pm
2e and 3p

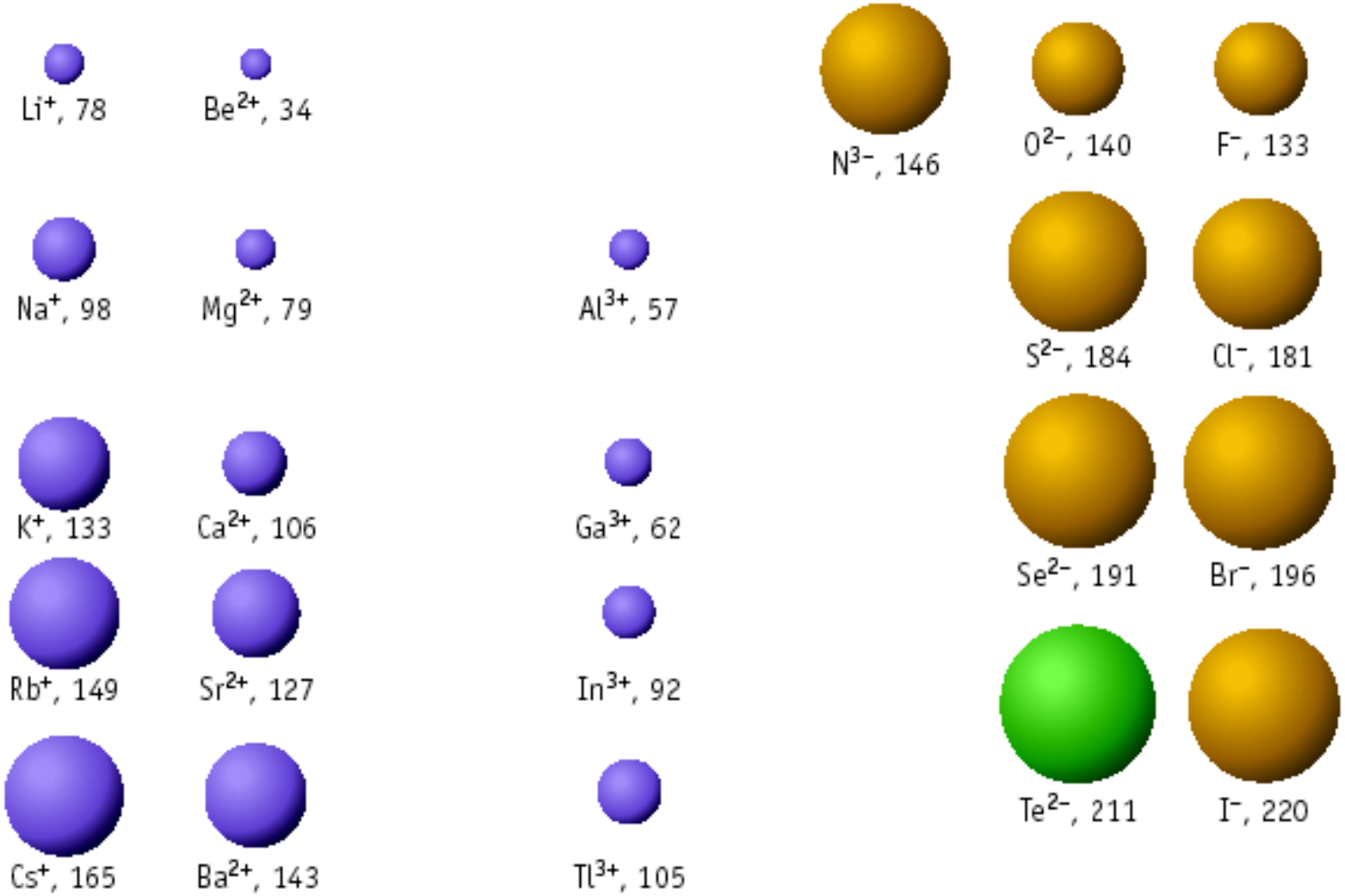
Forming a cation.

- When happens to atomic size when it gains electrons?
  - ANIONS are LARGER than the atoms from which they come.
  - The electron/proton attraction has gone DOWN and so size INCREASES.

Trends in Ion Sizes

Questions

- Why do metals lose electrons in their reactions?
- Why does Mg form Mg^{2+} ions, not Mg^{3+}?
- Why do nonmetals gain electrons?



Ionization Energy

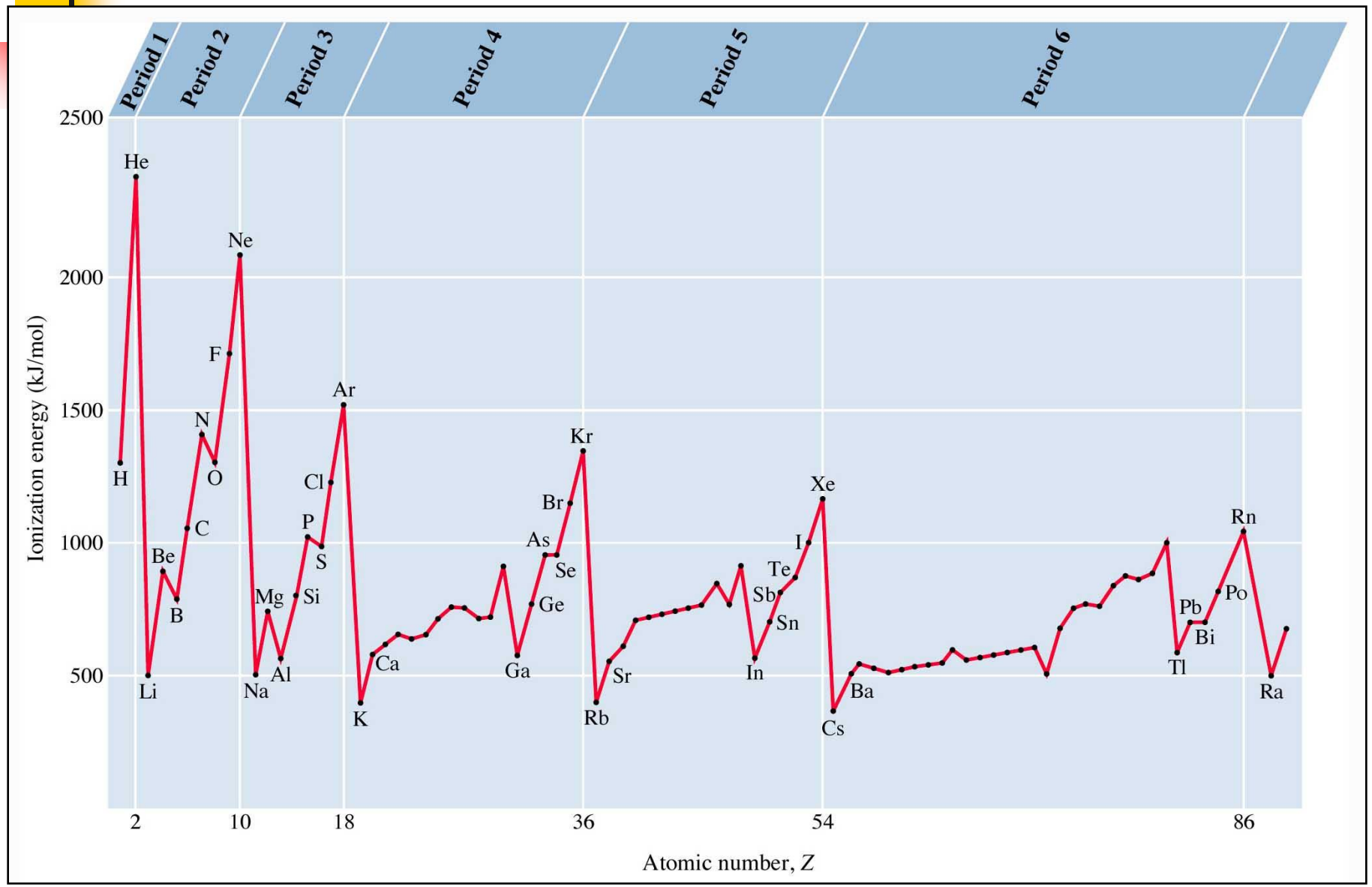
- **First ionization energy:** the minimal energy needed to remove the highest energy (outermost) electron from the neutral atom.
- For a lithium atom, the first ionization energy is illustrated by:

$$\mathbf{Li(1s^2 2s^1) \rightarrow Li^+(1s^2) + e^-}$$

Ionization energy = 520 kJ/mol

- There is a general trend that ionization energies increase with atomic number within a given period.
- This follows the trend in size, as it is more difficult to remove an electron that is closer to the nucleus.
- For the same reason, ionization energies, again following the trend in size, decrease as we descend a column of elements.

Ionization energy versus atomic number.



- The electrons of an atom can be removed successively.
- The energies required at each step are known as the ***first*** ionization energy, the ***second*** ionization energy, and so forth.

Table 8.3

Successive Ionization Energies of the First Ten Elements (kJ/mol)*

Element	First	Second	Third	Fourth	Fifth	Sixth	Seventh
H	1312						
He	2372	5250					
Li	520	7298	11,815				
Be	899	1757	14,848	21,006			
B	801	2427	3660	25,025	32,826		
C	1086	2353	4620	6222	37,829	47,276	
N	1402	2857	4578	7475	9445	53,265	64,358
O	1314	3388	5300	7469	10,989	13,326	71,333
F	1681	3374	6020	8407	11,022	15,164	17,867
Ne	2081	3952	6122	9370	12,177	15,238	19,998

*Ionization energies to the right of a vertical line correspond to removal of electrons from the core of the atom.

Factors that influence I.E.

- **Atomic size:** the larger the size of atom, the smaller the I.E.
- **Effective nuclear charge:** greater effective nuclear charge means more difficulty in electron removal from an atom due to increased attractive force between the nucleus and outermost electron, hence the greater the energy required to remove the electron.
- **Penetration effect of orbitals:** The order of energy required to remove electron from orbitals of a shell is s>p>d>¦.
- **Shielding or screening effect:** Screening effect reduces force of attraction between the nucleus and the outermost electron and lesser energy is required to separate the electron, which lowers I.E value.
- **Stability of half-filled and fully-filled orbitals:** Stability of half filled or completely filled degenerate orbitals is comparatively high (Hund's rule). So, electron separation from atoms needs more energy.

Electron Affinity

- The **electron affinity** is **the energy change for the process of adding an electron to a neutral atom in the gaseous state to form a negative ion**.
- For a chlorine atom, the first electron affinity is illustrated by:

$$Cl([Ne]3s^2 3p^5) + e^- \rightarrow Cl^-([Ne]3s^2 3p^6)$$

Electron Affinity = -349 kJ/mol

- More negative the electron affinity forms more stable -ive ion.
- Electron affirnity increases across a period; EA gets more -ive.
- Affinity decreases down a group (EA becomes less negative).
- Generally, the general trend goes from lower left to upper right as electron affinities become more negative.
- **ΔE is zero for N^- due to electron-electron repulsions.**

Electron affinity

- The more negative the electron affinity, the more stable the negative ion that is formed.
- Affinity for electron increases across a period (EA becomes more negative).
- Affinity decreases down a group (EA becomes less negative).
- Broadly speaking, the general trend goes from lower left to upper right as electron affinities become more negative.
- Electron affinity increases along the period while decreases down the group.

Table 8.4

Electron Affinities of the Main-Group Elements (kJ/mol)*

Period	IA	IIIA	IVA	VA	VIA	VIIA
1	H					
	−73					
2	Li	B	C	N	O	F
	−60	−27	−122	0	−141	−328
3	Na	Al	Si	P	S	Cl
	−53	−44	−134	−72	−200	−349
4	K	Ga	Ge	As	Se	Br
	−48	−30	−120	−77	−195	−325
5	Rb	In	Sn	Sb	Te	I
	−47	−30	−121	−101	−190	−295
6	Cs	Tl	Pb	Bi	Po	At
	−45	−30	−110	−110	−180	−270

*Atoms of the alkaline earth metals (Group IIA) and the noble gases (Group VIIIA) do not form stable negative ions.

Factors affecting the magnitude of electron affinity (EA)

- **Atomic size –** EA value decreases when atomic radius increases because electrostatic force of attraction decreases between the electron being added and the atomic nucleus due to increase of distance between them.
- **Effective nuclear charge** – the more the effective nuclear charge, the stronger the electrostatic force between nucleus and added electron and the stronger the EA.
- **Screening or Shielding effect** – Electron affinity of elements decreases with the increasing shielding or screening effect and this arises when number of electrons in the inner shell increases.
- **Stability of half filled and completely filled orbitals** – The stability of half filled and completely filled degenerate orbitals of a sub shell is comparatively more, so it is difficult to add electron in such orbitals and lesser energy is released on addition of electron hence the electron affinity value will decrease.